1. Setting

Let $A \subseteq \mathbb{R}^{d_A}$ be the arm set and let $D \subseteq \mathbb{R}^{d_D}$ be the outcome set. We assume that A and D are Euclidean balls of radii R_A and R_D (not necessarily centred at the origin). An element $x \in A$ is called an **arm**, while an element $y \in D$ is a possible sampled outcome. The reward is affine in the arm and outcome:

$$r(x, y) := a^T x + b^T y + c.$$

Let $Z \subseteq \mathbb{R}^{d_Z}$ denote the hypothesis set, and let $z^\star \in Z$ be the unknown true hypothesis. For each arm $x \in A$, define the true feasible outcome set

$$K^\star(x) := \{y \in D : F_0(x, z^\star) + F_1(y, z^\star) = 0\},$$

where F_0 and F_1 are both bilinear maps.

After fixing $z^\star$, the constraint can be written in intrinsic affine coordinates as

$$Cy + Bx + d = 0,$$

where B, C, and d are fixed once $z^\star$ is fixed. Equivalently,

$$K^\star(x) = \{y \in D : Cy + Bx + d = 0\}. \tag{1}$$

We assume throughout that $K^\star(x)$ is nonempty for every $x \in A$.

The interaction proceeds for rounds $t = 1, ..., T$. Let $\mathcal{F}_{t-1}$ denote the history before round t. The learner chooses an arm $x_t \in A$. Nature may then choose, adaptively as a function of the history and x_t, any distribution supported on D whose conditional mean

$$m_t := E[y_t \mid \mathcal{F}_{t-1}, x_t]$$

belongs to $K^\star(x_t)$. An outcome y_t is sampled from this distribution, and the learner receives reward $r(x_t, y_t)$.

The robust value of an arm and the optimal robust value are

$$v^\star(x) := \min_{y \in K^\star(x)} r(x, y), \quad V^\star := \max_{x \in A} v^\star(x).$$

The regret over horizon T is

$$R_T := TV^\star - \sum_{t=1}^{T} r(x_t, y_t).$$

Finally, let

$$C_r := \max_{x \in A, y \in D} r(x, y) - \min_{x \in A, y \in D} r(x, y)$$

denote the reward range.

Theorem 1 (efficient $T^{\frac{8}{9}}$ learning). *Suppose that the known problem data in the setting above are rational. For every known horizon T, there is a policy whose arithmetic and weak-optimization running time is polynomial in T and the input bit length and which, for every true hypothesis and every compatible adaptive nature policy, satisfies*

$$E[R_T] \leq \tilde{O}\Big(P(d_A, d_D, R_A, R_D, \|b\|_2, C_r) T^{\frac{8}{9}}\Big),$$

where P is a fixed polynomial. No geometric property of Z is used beyond its inducing a valid family of compatible sets of the form Equation 1.

2. Warmup: Noiseless Setting

In this section, playing an arm x reveals an exact feasible response $y \in K^\star(x)$; there is no sampling noise. However, there still is the Knightian uncertainty corresponding to the adversary picking an arbitrary outcome inside $K^\star(x)$.

Lemma 1. *There exist an affine map $f : A \to \mathbb{R}^{d_D}$ and a fixed linear subspace $N \subseteq \mathbb{R}^{d_D}$ such that*

$$K^\star(x) = (f(x) + N) \cap D, \quad x \in A.$$

Equivalently, before intersecting with D, the feasible sets form a family of parallel affine spaces.

Lemma 2. *Suppose that, for every arm $x \in A$, we have identified a nonempty set $K'(x) \subseteq K^\star(x)$, and suppose that whenever the learner plays x, nature chooses an outcome in $K'(x)$. Define*

$$v'(x) := \min_{y \in K'(x)} r(x, y),$$

and choose

$$x' \in \operatorname{argmax}_{x \in A} v'(x).$$

Then playing x' on every round incurs nonpositive regret with respect to the original robust benchmark, i.e. $R_T \le 0$.

Lemma 3. *There exist $d_A + 1$ arms $x^{(0)}, ..., x^{(d_A)} \in A$ such that, from any possible sequence of responses $y^{(i)} \in K^\star\big(x^{(i)}\big)$, we can compute an affine map $\hat{f}$ such that*

$$K^\star(x) = \Big(\hat{f}(x) + N\Big) \cap D, \quad x \in A.$$

Proof. Let $x^{(0)}, ..., x^{(d_A)}$ be any affine basis of A. Play them in sequence and let $y^{(0)}, ..., y^{(d_A)}$ be the corresponding outcomes chosen by the adversary. Let $\hat{f}$ be the unique affine interpolator for which $\hat{f}\big(x^{(i)}\big) = y^{(i)}$ for all i. Since $y^{(i)} \in K^\star\big(x^{(i)}\big)$, we have that $\hat{f}\big(x^{(i)}\big) + N = f\big(x^{(i)}\big) + N$ for all i. Since $x^{(i)}$'s form an affine basis, therefore $\hat{f}(x) + N = f(x) + N$ for all $x \in A$. ■

Now the learner works as follows.